

SHARDARA, Kazakhstan

WMO: 384390

Lat: 41.2492N

Lon: 67.9692E

Elev: 902

StdP: 14.22

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	12.7	2.6	6.5	7.3	8.3	8.7	13.6	21.0	39.2	18.9	39.0	4.7	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	101.0	70.3	98.1	69.4	95.6	68.7	75.3	93.5	73.3	91.8	71.6	90.5	7.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	110.6	88.5	66.7	101.7	86.6	64.7	94.5	84.4	39.5	93.4	37.6	91.9	36.1	90.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.6	15.7	13.5	DB	4.5	106.7	7.8	2.4	-1.0	108.4	-5.6	109.8	-9.9	111.2	-15.6	113.0
			WB	3.8	79.0	7.6	2.6	-1.6	80.9	-6.1	82.4	-10.3	83.9	-15.8	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	32.4	36.0	48.3	60.4	72.0	80.9	84.3	81.7	71.3	58.4	45.3	34.2
	DBStd	20.10	9.24	10.15	8.47	8.19	6.32	5.31	4.59	5.28	6.89	8.55	9.43	8.93
	HDD50	1802	547	395	133	15	0	0	0	0	0	27	193	492
	HDD65	4342	1012	811	518	181	15	0	0	0	22	235	592	956
	CDD50	5052	0	3	81	327	684	927	1064	984	640	288	53	1
	CDD65	2115	0	0	0	42	233	477	599	519	212	31	2	0
	CDH74	25541	0	0	3	309	2184	5945	8222	6534	2112	227	5	0
	CDH80	13097	0	0	0	70	789	3075	4746	3559	816	42	0	0
Wind	WSAvg	5.9	5.6	5.9	5.8	5.8	6.5	6.4	6.3	6.1	5.7	5.2	5.7	5.8
Precipitation	PrecAvg	16.1	1.9	2.3	2.4	2.2	1.5	0.5	0.2	0.1	0.2	0.9	1.8	2.0
	PrecMax	23.5	3.0	5.1	4.1	5.2	3.0	1.4	1.0	0.5	0.7	3.0	5.0	4.9
	PrecMin	8.5	0.7	0.2	0.7	0.5	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2
	PrecStd	3.4	0.7	1.1	0.9	1.1	0.9	0.4	0.3	0.1	0.2	0.7	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	60.9	72.9	86.2	94.4	102.3	105.8	103.9	95.5	84.8	73.1	55.4
		MCWB	45.4	51.3	57.3	64.2	66.2	69.5	71.6	72.4	68.2	61.9	57.8	47.1
	2%	DB	49.3	55.1	68.3	81.7	90.5	98.2	101.8	99.8	91.3	79.7	66.8	51.8
		MCWB	43.2	46.9	54.6	61.9	65.8	69.3	70.2	70.4	66.3	59.9	55.0	45.4
	5%	DB	45.9	51.2	64.4	77.4	87.6	95.5	99.0	97.0	88.1	75.8	62.4	48.2
		MCWB	41.2	44.8	53.6	61.2	65.0	68.2	69.6	69.4	65.2	58.5	52.4	42.6
	10%	DB	43.2	48.0	60.6	73.4	84.5	92.9	96.1	94.3	84.8	72.3	58.3	45.3
		MCWB	39.5	43.0	51.9	60.0	64.4	67.5	68.9	68.7	64.0	56.8	50.1	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.3	51.3	59.6	67.3	71.4	76.3	78.0	78.8	72.4	64.3	59.1	49.1
		MCDB	52.4	60.0	68.7	80.0	87.0	93.8	95.8	95.4	89.5	80.6	70.3	53.8
	2%	WB	43.8	47.8	56.7	64.9	69.0	73.5	75.2	75.2	69.2	61.6	55.9	45.6
		MCDB	48.4	53.7	65.2	76.4	83.7	91.0	94.2	93.1	86.5	76.3	65.2	50.1
	5%	WB	41.6	45.4	54.5	62.8	67.4	71.6	73.4	72.9	66.9	59.7	53.1	43.2
		MCDB	45.4	50.4	62.7	74.9	82.5	89.2	92.8	91.1	85.0	73.5	61.3	47.5
	10%	WB	39.7	43.2	52.3	60.7	66.0	70.0	71.7	70.8	64.8	57.8	50.7	41.2
		MCDB	43.1	47.7	59.8	71.9	80.9	87.6	91.7	90.0	83.0	70.9	57.6	45.1
Mean Daily Temperature Range	5% DB	MDBR	11.0	13.8	16.2	18.1	20.2	21.8	22.9	23.8	23.9	20.6	15.1	10.8
		MCDBR	13.6	16.9	20.3	23.4	23.9	24.8	26.4	26.1	25.2	23.6	18.4	13.6
		MCWBR	9.6	11.1	11.2	11.0	10.7	10.6	11.3	12.5	12.4	12.0	10.7	9.4
	5% WB	MCDBR	12.7	15.4	17.6	20.2	20.5	21.0	22.2	23.5	24.6	21.9	16.6	12.7
		MCWBR	9.4	10.9	10.7	11.3	11.0	11.6	12.8	14.2	14.4	12.2	10.5	9.2
Clear-Sky Solar Irradiance	taub	0.388	0.427	0.480	0.497	0.480	0.469	0.446	0.422	0.412	0.466	0.449	0.389	
	taud	2.070	1.991	1.913	1.921	2.007	2.045	2.117	2.164	2.131	1.928	1.955	2.042	
	Ebn at Noon	227	238	240	247	255	258	263	266	259	220	201	213	
	Edh at Noon	36	45	55	58	55	53	49	45	43	48	40	34	
All-Sky Solar Radiation	RadAvg	532	816	1218	1697	2156	2435	2420	2175	1729	1139	680	477	
	RadStd	63	93	100	166	110	92	77	62	54	79	85	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)			N/A	N/A	N/A	+1.03	N/A	N/A	N/A	+185	+139

Nomenclature: See separate page